

# TD Autosélection - Corrigé

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**Definition: Self-selection.** Which individuals choose themselves into different groups, options or contracts according to their characteristics, preferences or information. Their choices can therefore reveal information about their characteristics that are not observable for an outside party.

## 1 Regulation of a monopoly (Baron-Myerson 1982)

We are interested in the optimization problem of a social planner, which maximizes the welfare of consumers by regulating a monopoly firm, under the constraint of the firm's costs. Let  $C(q, \theta) = \theta q$  be the cost function of the monopoly which produces quantity  $q$ .

Let  $\theta$  be the type of the monopoly.  $\theta = \theta_L$  with probability  $\beta$ , and  $\theta = \theta_H > \theta_L$  with probability  $1 - \beta$ .

The regulator delegates the production of a good to a monopoly: the monopoly receives a monetary transfer  $t$  from the regulator, so that the profit of the monopolist is  $(t - \theta q)$ . The monopolist agrees to produce only if the regulator offers them a non-negative profit. The consumer surplus for  $q$  units is equal to  $S(q)$  with  $S' > 0$ ,  $S'' < 0$  and  $S(0) = 0$ .

**Question 0:** Who is the principal? Who is the agent?

The principal is the uninformed party who designs the contract: here, the regulator. The agent is the informed party who chooses whether to accept the contract: here, the monopolist.

**Question 1:** Suppose that the regulator can observe the type  $\theta$  of the monopolist. The regulator proposes a contract  $(q_L, t_L)$  to the monopolist of type  $\theta_L$ , and a contract  $(q_H, t_H)$  to a monopolist of type  $\theta_H$ . The monopolist has the choice to accept the contract or refuse the contract and not produce anything. What are the quantities  $q_L^*$  and  $q_H^*$  that maximize the welfare of consumers? What are the transfers  $t_L^*$  and  $t_H^*$  that ensure participation of the monopolist?

Fix a type  $i \in \{L, H\}$ . Because the regulator observes  $\theta_i$ , it can solve the problem type by type:

$$\max_{q_i, t_i} S(q_i) - t_i \quad \text{subject to} \quad t_i - \theta_i q_i \geq 0.$$

The constraint is the firm's participation constraint. Refusing the contract yields zero profit, so the contract must give at least zero.

The participation constraint must bind. If  $t_i - \theta_i q_i > 0$ , the regulator could reduce  $t_i$  slightly,

keep  $q_i$  unchanged, still satisfy participation, and improve the objective. Hence

$$t_i - \theta_i q_i = 0 \quad \implies \quad t_i = \theta_i q_i.$$

Substituting into the objective gives

$$\max_{q_i} S(q_i) - \theta_i q_i.$$

The first-order condition is

$$S'(q_i) - \theta_i = 0,$$

so the efficient quantity satisfies

$$S'(q_i^*) = \theta_i.$$

Because  $S'' < 0$ , the objective is concave and this gives the unique optimum. Therefore

$$S'(q_L^*) = \theta_L, \quad S'(q_H^*) = \theta_H,$$

and the transfers ensuring participation are

$$t_L^* = \theta_L q_L^*, \quad t_H^* = \theta_H q_H^*.$$

Economically, the efficient quantity equates marginal consumer benefit  $S'(q)$  and marginal production cost  $\theta_i$ .

**Question 2:** Suppose now that the regulator does not observe the type of the monopolist. The regulator proposes a menu of contracts to the monopoly:  $(q_L, t_L)$  and  $(q_H, t_H)$ . The monopolist has to choose between the two contracts and produce (or refuse both contracts and not produce). Show that if the regulator proposes the contracts  $(q_L^*, t_L^*)$  and  $(q_H^*, t_H^*)$ , one of the two types has an incentive to pretend to be of the other type.

Under full information, both participation constraints bind, so each type earns zero on its own contract. Consider a type- $L$  firm choosing the contract intended for type  $H$ . It receives  $t_H^* = \theta_H q_H^*$ , but its true marginal cost is still  $\theta_L$ . Its profit from mimicking is therefore

$$t_H^* - \theta_L q_H^* = \theta_H q_H^* - \theta_L q_H^* = (\theta_H - \theta_L) q_H^* > 0.$$

Hence type  $L$  strictly prefers the  $H$ -contract to its own zero-rent contract. The full information allocation is therefore not incentive compatible when the type is private.

**Question 3:** We are now looking for the second best optimum, the menu  $(q_L^{**}, t_L^{**}), (q_H^{**}, t_H^{**})$  that maximizes the welfare of consumers knowing that the regulator cannot observe the type of the monopolist. Write down the incentive constraints for a monopolist of type  $H$  (ICH) and of type  $L$  (ICL), as well as their participation constraints. What is the regulator's optimization problem now?

Because the regulator no longer observes type, a firm can choose either contract. When a type mimics the other type, the chosen contract changes but the true cost parameter does not.

For type  $L$ , truthful choice gives

$$\pi_L(L) = t_L - \theta_L q_L,$$

whereas choosing the  $H$ -contract gives

$$\pi_L(H) = t_H - \theta_L q_H.$$

Truth-telling therefore requires

$$t_L - \theta_L q_L \geq t_H - \theta_L q_H \quad (ICL).$$

(truthful telling yields a higher utility than pretending to be the other type) For type  $H$ , truthful choice gives

$$\pi_H(H) = t_H - \theta_H q_H,$$

and mimicking type  $L$  gives

$$\pi_H(L) = t_L - \theta_H q_L.$$

Hence

$$t_H - \theta_H q_H \geq t_L - \theta_H q_L \quad (ICH).$$

Participation constraint: monopolist compares accepting contract and refusing it (which yields zero)

$$t_L - \theta_L q_L \geq 0 \quad (IRL),$$

$$t_H - \theta_H q_H \geq 0 \quad (IRH).$$

Thus the regulator solves

$$\max_{(t_L, q_L), (t_H, q_H)} \beta (S(q_L) - t_L) + (1 - \beta) (S(q_H) - t_H)$$

s.t.  $ICL$ ,  $ICH$ ,  $IRL$ , and  $IRH$ .

IR (individual rationality) is the participation constraint, while IC is the incentive constraint

**Question 4:** Show that the constraint  $IRL$  is automatically satisfied if the other constraints are satisfied. Show that  $ICL$  is binding in the optimum.

Start from  $ICL$ :

$$t_L - \theta_L q_L \geq t_H - \theta_L q_H.$$

Since  $\theta_H > \theta_L$  and  $q_H \geq 0$ ,

$$t_H - \theta_L q_H \geq t_H - \theta_H q_H.$$

Finally,  $IRH$  gives

$$t_H - \theta_H q_H \geq 0.$$

Combining the three inequalities,

$$t_L - \theta_L q_L \geq t_H - \theta_L q_H \geq t_H - \theta_H q_H \geq 0.$$

Therefore *IRL* is implied by *ICL* and *IRH*: it is redundant.

Now suppose that *ICL* were slack:

$$t_L - \theta_L q_L > t_H - \theta_L q_H.$$

Reduce  $t_L$  by a sufficiently small  $\varepsilon > 0$ . Then:

- *IRH* is unchanged because  $t_H$  is unchanged;
- *ICH* becomes easier to satisfy because its right-hand side contains  $t_L$ , which falls;
- *ICL* remains satisfied for sufficiently small  $\varepsilon$ , because it was initially strict;
- *IRL* remains satisfied because it is implied by *ICL* and *IRH*.

The regulator's objective contains  $-\beta t_L$ , so lowering  $t_L$  strictly raises expected consumer welfare. Hence a slack *ICL* cannot be optimal. The regulator pushes  $t_L$  down until type  $L$  is exactly indifferent between its own contract and the  $H$ -contract:

$$t_L - \theta_L q_L = t_H - \theta_L q_H.$$

Thus *ICL* binds at the optimum.

**Question 5:** By combining *ICL* and *ICH*, show that  $q_L \geq q_H$ . Under this constraint, show next that the *ICH* can be ignored. Show that the *IRH* is binding in the optimum.

Add the two incentive constraints:

$$(t_L - \theta_L q_L) + (t_H - \theta_H q_H) \geq (t_H - \theta_L q_H) + (t_L - \theta_H q_L).$$

The transfers cancel, leaving

$$-\theta_L q_L - \theta_H q_H \geq -\theta_L q_H - \theta_H q_L.$$

Rearranging,

$$(\theta_H - \theta_L) q_L \geq (\theta_H - \theta_L) q_H.$$

Since  $\theta_H - \theta_L > 0$ , we obtain

$$q_L \geq q_H.$$

From Question 4, *ICL* binds:

$$t_L - \theta_L q_L = t_H - \theta_L q_H,$$

so

$$t_L - t_H = \theta_L (q_L - q_H).$$

Because  $q_L - q_H \geq 0$  and  $\theta_H \geq \theta_L$ ,

$$\theta_L(q_L - q_H) \leq \theta_H(q_L - q_H).$$

Therefore

$$\begin{aligned} t_L - t_H &\leq \theta_H(q_L - q_H) \\ \iff t_H - \theta_H q_H &\geq t_L - \theta_H q_L. \end{aligned}$$

This is exactly *ICH*. Hence, once monotonicity  $q_L \geq q_H$  and binding *ICL* are imposed, *ICH* is automatically satisfied.

It remains to show that *IRH* binds. Suppose instead that the high type gets strictly positive participation rent:

$$t_H - \theta_H q_H > 0.$$

Reduce both transfers by the same small amount  $\varepsilon > 0$ :

$$t'_L = t_L - \varepsilon, \quad t'_H = t_H - \varepsilon.$$

Because both transfers fall by the same amount, utility differences between contracts are unchanged, so both incentive constraints are unchanged. Since *IRH* was slack, it remains satisfied for sufficiently small  $\varepsilon$ ; and then *IRL* also remains satisfied because it is implied by *ICL* and *IRH*. Quantities are unchanged, but expected transfers fall by  $\varepsilon$  in every state, so the regulator's objective strictly increases. This contradicts optimality. Therefore

$$t_H - \theta_H q_H = 0,$$

i.e. *IRH* binds.

**Question 6:** Solve the now simplified optimization problem of the regulator. What are the quantities produced  $q_L^{**}, q_H^{**}$  in the second best optimum?

The two binding constraints are now

$$IRH : \quad t_H = \theta_H q_H,$$

and

$$ICL : \quad t_L - \theta_L q_L = t_H - \theta_L q_H.$$

Substituting the first equation into the second gives

$$t_L - \theta_L q_L = \theta_H q_H - \theta_L q_H = (\theta_H - \theta_L) q_H,$$

so

$$t_L = \theta_L q_L + (\theta_H - \theta_L) q_H, \quad t_H = \theta_H q_H.$$

The regulator's objective becomes

$$\begin{aligned} W(q_L, q_H) &= \beta(S(q_L) - t_L) + (1 - \beta)(S(q_H) - t_H) \\ &= \beta(S(q_L) - \theta_L q_L - (\theta_H - \theta_L)q_H) + (1 - \beta)(S(q_H) - \theta_H q_H) \\ &= \beta(S(q_L) - \theta_L q_L) + (1 - \beta)(S(q_H) - \theta_H q_H) - \beta q_H(\theta_H - \theta_L). \end{aligned}$$

For  $q_L$ ,

$$\frac{\partial W}{\partial q_L} = \beta(S'(q_L) - \theta_L),$$

so the first-order condition is

$$S'(q_L^{**}) = \theta_L.$$

Under full information we also had  $S'(q_L^*) = \theta_L$ , so strict concavity implies

$$q_L^{**} = q_L^*.$$

There is therefore no distortion for the efficient type.

For  $q_H$ ,

$$\frac{\partial W}{\partial q_H} = (1 - \beta)(S'(q_H) - \theta_H) - \beta(\theta_H - \theta_L).$$

Setting this equal to zero gives FOC:

$$(1 - \beta)S'(q_H^{**}) = (1 - \beta)\theta_H + \beta(\theta_H - \theta_L),$$

Isolate  $S'$ :

$$S'(q_H^{**}) = \theta_H + \frac{\beta}{1 - \beta}(\theta_H - \theta_L).$$

The extra term is strictly positive since  $\beta > 0$  and  $\theta_H > \theta_L$ , hence

$$S'(q_H^{**}) > \theta_H = S'(q_H^*).$$

Since  $S'' < 0$ ,  $S'$  is decreasing, so this implies

$$q_H^{**} < q_H^*.$$

The high-cost type therefore produces less in the second-best optimum than in the first-best.

Finally, verify monotonicity. Because  $S'$  is decreasing,

$$q_L^{**} \geq q_H^{**} \iff S'(q_L^{**}) \leq S'(q_H^{**}) \iff \theta_L \leq \theta_H + \frac{\beta}{1 - \beta}(\theta_H - \theta_L),$$

which holds since  $\theta_L \leq \theta_H$ .

**Question 7:** Show that a monopolist of type  $\theta_L$  receives an “information rent” (i.e. their profits are strictly positive). Why do we talk about a rent extraction / efficiency trade-off? Comment.

The low type's profit is

$$\pi_L = t_L^{**} - \theta_L q_L^{**}.$$

Using the transfer formula found above,

$$t_L^{**} = \theta_L q_L^{**} + (\theta_H - \theta_L) q_H^{**},$$

we obtain

$$\pi_L = (\theta_H - \theta_L) q_H^{**}.$$

Since  $\theta_H > \theta_L$  and the interior solution has  $q_H^{**} > 0$ ,

$$\pi_L = (\theta_H - \theta_L) q_H^{**} > 0.$$

This is the information rent.

Why must the regulator leave this rent? If type  $L$  chose the  $H$ -contract, its profit would be

$$t_H - \theta_L q_H = \theta_H q_H - \theta_L q_H = (\theta_H - \theta_L) q_H.$$

To prevent this deviation, the contract intended for type  $L$  must give at least the same utility. The rent therefore rises with  $q_H$ . The regulator can reduce the rent by reducing  $q_H$ , but that makes the high-cost type produce less than the second-best efficient quantity. Hence we have a rent extraction / efficiency trade-off: extracting more rent requires a stronger distortion of the inefficient type's allocation.

## 2 Discriminating Monopoly (Mussa-Rosen 1978)

**Definition: Discriminating monopoly.** A monopoly that offers different prices, quantities or contract terms to different consumers, depending on their preferences, characteristics or willingness to pay.

A monopoly firm sells goods to heterogeneous consumers. If a consumer purchases  $q$  units of the good and pays the total sum  $T$  for them, their utility can be described as:

$$u(q, T, \theta) = \theta q - T$$

where  $\theta$  is the “type” of the consumer, which is their private information. Suppose that  $\theta$  is continuously distributed over the support  $[\underline{\theta}, \bar{\theta}]$  with a probability density function  $f$  and a cumulative density function  $F$ . The profit of the monopoly, which sells  $q$  units at a price of  $T$  is

$$\pi = T - \frac{q^2}{2}.$$

In order to maximize their profit, the monopoly chooses a “menu” of contracts  $(q(\theta), t(\theta))$  which are designed so that a consumer of type  $\theta$  chooses the contract that is designed for them (incentive constraint) and so that they obtain at least their reservation utility (participation constraint). A menu of contracts is a pair of functions  $(q(\cdot), t(\cdot))$ , defined on  $[\underline{\theta}, \bar{\theta}]$  with values in  $\mathbb{R}_+$ .

**Question 1:** Suppose the monopolist can observe  $\theta$ . Characterize the solutions  $(q^*(\theta), t^*(\theta))$ , and comment.

For a given type  $\theta$ , the monopolist solves

$$\max_{q,T} T - \frac{q^2}{2} \quad \text{s.t.} \quad \theta q - T \geq 0.$$

The participation constraint must bind. If  $\theta q - T > 0$ , the monopolist can increase  $T$  slightly, while still leaving the consumer willing to participate while increasing profit. Thus

$$T - \theta q = 0,$$

which gives

$$T = \theta q.$$

Substituting into profit maximizing objective gives

$$\max_q \theta q - \frac{q^2}{2}.$$

The first-order condition is

$$\theta - q = 0,$$

so

$$q^*(\theta) = \theta.$$

The second derivative is  $-1 < 0$ , the function is strictly concave, so this is the unique optimum. The corresponding payment is

$$T^*(\theta) = \theta q^*(\theta) = \theta^2.$$

The consumer gets zero utility (monopolist squeezes the entire surplus):

$$U^*(\theta) = \theta q^*(\theta) - T^*(\theta) = 0.$$

Total surplus is

$$S(q, \theta) = \theta q - \frac{q^2}{2},$$

which has exactly the same FOC as profit maximizing because the monopoly gets the total surplus, so the solution  $q^*(\theta) = \theta$  is efficient. Optimal profit is

$$\pi^*(\theta) = \theta^2 - \frac{\theta^2}{2} = \frac{\theta^2}{2}.$$

Suppose now that the monopolist does not observe  $\theta$ . The monopolist knows that  $\theta$  is continuously distributed over  $[\underline{\theta}, \bar{\theta}]$  with a pdf  $f$  and a cdf  $F$ .

**Question 2:** Show that the incentive constraints imply that  $q(\theta)$  is increasing.

(Tip: Write the incentive constraints for a pair of types  $\theta \neq \theta'$ .)

Take two types  $\theta$  and  $\theta'$ . Incentive constraints require

$$\theta q(\theta) - T(\theta) \geq \theta q(\theta') - T(\theta'),$$



and

$$\theta'q(\theta') - T(\theta') \geq \theta'q(\theta) - T(\theta).$$

Which means that the utility from being truthful must be at least equal to the utility from pretending to be the other type.

Add the two inequalities.

$$\theta q(\theta) - T(\theta) + \theta'q(\theta') - T(\theta') \geq \theta q(\theta') - T(\theta') + \theta'q(\theta) - T(\theta).$$

The transfer terms cancel:

$$\theta q(\theta) + \theta'q(\theta') \geq \theta q(\theta') + \theta'q(\theta).$$

Rearranging we get

$$(\theta - \theta')(q(\theta) - q(\theta')) \geq 0.$$

If  $\theta > \theta'$ , the first factor is positive, so the second must be non-negative:

$$q(\theta) \geq q(\theta').$$

Hence  $q(\theta)$  is nondecreasing.

**Question 3:** In the following, since  $U(\theta) = \theta q(\theta) - T(\theta)$ , the variable of interest is the consumer surplus  $U(\theta)$  rather than  $T(\theta)$ . Show that given the incentive and participation constraints:

$$U(\theta) = \int_{\underline{\theta}}^{\theta} q(x) dx. \quad (1)$$

(Tip: Write the incentive constraints between  $\hat{\theta}$  and  $\theta \neq \hat{\theta}$ . We interpret the utility at  $\theta$  as a maximum, in order to have a relationship between  $q'(\cdot)$  and  $T'(\cdot)$ .)

Let  $V(\theta, \hat{\theta}) = \theta q(\hat{\theta}) - T(\hat{\theta})$  be  $\theta$ 's utility from pretending to be  $\hat{\theta}$ , and let  $U(\theta) = V(\theta, \theta)$  be  $\theta$ 's utility from telling the truth.

#### Incentive constraint

Truthful reporting must maximize the consumer's utility:

$$\theta \in \arg \max_{\hat{\theta}} V(\theta, \hat{\theta}).$$

Thus, at optimum, the FOC is

$$\left. \frac{\partial V(\theta, \hat{\theta})}{\partial \hat{\theta}} \right|_{\hat{\theta}=\theta} = 0.$$

We have

$$U(\theta) = V(\theta, \theta).$$

$\theta$  appears in both arguments of  $V$ : as the true type and as the truthful report, which means

that we have to use the chain rule:

$$U'(\theta) = \partial_1 V(\theta, \theta) + \partial_2 V(\theta, \theta)$$

From the incentive constraint, second term is equal to 0 (can also be deduced from the envelope):

$$\partial_2 V(\theta, \theta) = \theta q'(\theta) - T'(\theta) = 0.$$

Here,  $\partial_1 V$  and  $\partial_2 V$  denote the partial derivatives with respect to the first and second arguments of  $V$ , respectively. For the first term, differentiating yields

$$\partial_1 V(\theta, \theta) = q(\theta).$$

Therefore,

$$U'(\theta) = \underbrace{\partial_1 V(\theta, \theta)}_{=q(\theta)} + \underbrace{\partial_2 V(\theta, \theta)}_{=0} = q(\theta).$$

Integrating between  $\underline{\theta}$  and  $\theta$ :

$$\int_{\underline{\theta}}^{\theta} U'(x) dx = \int_{\underline{\theta}}^{\theta} q(x) dx.$$

We have

$$U(\theta) - U(\underline{\theta}) = \int_{\underline{\theta}}^{\theta} q(x) dx,$$

**Participation constraint** Participation requires  $U(\theta) \geq 0$  for all  $\theta$ . Since

$$U'(\theta) = q(\theta) \geq 0,$$

utility is nondecreasing in type. It is sufficient to impose participation for the lowest type:

$$U(\underline{\theta}) \geq 0.$$

At the optimum, this constraint binds. Otherwise, the monopolist could decrease the utility of the lowest type leading to an increase in its profit without compromising the participation of the lowest type.

Therefore,

$$U(\underline{\theta}) = 0.$$

Which yields, finally,

$$U(\theta) = \int_{\underline{\theta}}^{\theta} q(x) dx.$$

**Question 4:** Write the optimization problem of the monopolist as a function of  $(q(\theta), U(\theta))$  and show that it simplifies to:

$$\max_{(q(\cdot), U(\cdot))} \int_{\underline{\theta}}^{\bar{\theta}} \left( \theta q(\theta) - U(\theta) - \frac{q(\theta)^2}{2} \right) f(\theta) d\theta \quad (2)$$

with  $U(\theta) = \int_{\underline{\theta}}^{\theta} q(x) dx$ .

Expected profit is

$$\max_{q(\cdot), T(\cdot)} \int_{\underline{\theta}}^{\bar{\theta}} \left( T(\theta) - \frac{q(\theta)^2}{2} \right) f(\theta) d\theta.$$

From Question 3, incentive and participation constraints imply

$$U'(\theta) = q(\theta), \quad U(\underline{\theta}) = 0,$$

therefore

$$U(\theta) = \int_{\underline{\theta}}^{\theta} q(x) dx.$$

Since

$$U(\theta) = \theta q(\theta) - T(\theta),$$

we solve for the transfer:

$$T(\theta) = \theta q(\theta) - U(\theta).$$

We can then substitute this into expected profit, which gives

$$\max_{q(\cdot), U(\cdot)} \int_{\underline{\theta}}^{\bar{\theta}} \left( \theta q(\theta) - U(\theta) - \frac{q(\theta)^2}{2} \right) f(\theta) d\theta,$$

s.t. the constraint

$$U(\theta) = \int_{\underline{\theta}}^{\theta} q(x) dx.$$

(also s.t. the constraint that  $q$  is increasing) (Notice the term  $-U(\theta)$  compared to the full information case: it is the cost of the information rent that must be left to consumers by the monopoly in order to obtain truthful revelation.)

**Question 5:** Deduce the quantities produced  $q^{**}(\theta)$  at the second best optimum. Suppose  $f(\theta) \neq 0$ , i.e. there exist consumers of each type over the support.

(Tip: Integrate by parts  $\int_{\underline{\theta}}^{\bar{\theta}} U(\theta) f(\theta) d\theta$  in (2).)

We only need to rewrite the middle term

$$\int_{\underline{\theta}}^{\bar{\theta}} U(\theta) f(\theta) d\theta.$$

Using Question 3,

$$U(\theta) = \int_{\underline{\theta}}^{\theta} q(x) dx,$$

so

$$\int_{\underline{\theta}}^{\bar{\theta}} \left( \int_{\underline{\theta}}^{\theta} q(x) dx \right) f(\theta) d\theta.$$

We now compute this term by integration by parts. Take

$$u(\theta) = \int_{\underline{\theta}}^{\theta} q(x) dx, \quad v'(\theta) = f(\theta),$$

therefore, we have

$$u'(\theta) = q(\theta), \quad v(\theta) = F(\theta).$$

Then

$$\begin{aligned} \left( \left( \int_{\underline{\theta}}^{\theta} q(x) dx \right) F(\theta) \right)_{\underline{\theta}}^{\bar{\theta}} - \int_{\underline{\theta}}^{\bar{\theta}} q(\theta) F(\theta) d\theta \\ = \left( \int_{\underline{\theta}}^{\bar{\theta}} q(x) dx \right) F(\bar{\theta}) - 0 - \int_{\underline{\theta}}^{\bar{\theta}} q(\theta) F(\theta) d\theta. \end{aligned}$$

Since  $F(\bar{\theta}) = 1$ , this becomes

$$\int_{\underline{\theta}}^{\bar{\theta}} q(\theta) d\theta - \int_{\underline{\theta}}^{\bar{\theta}} q(\theta) F(\theta) d\theta = \int_{\underline{\theta}}^{\bar{\theta}} q(\theta) (1 - F(\theta)) d\theta.$$

Equivalently,

$$\int_{\underline{\theta}}^{\bar{\theta}} \frac{1 - F(\theta)}{f(\theta)} q(\theta) f(\theta) d\theta.$$

Therefore the monopolist's problem becomes

$$\max_{q(\cdot)} \int_{\underline{\theta}}^{\bar{\theta}} \left( \theta q(\theta) - \frac{q(\theta)^2}{2} - q(\theta) \frac{1 - F(\theta)}{f(\theta)} \right) f(\theta) d\theta,$$

s.t.  $q$  increasing.

The strategy is now to suppose that  $q$  is increasing and solve under that assumption. Since the objective is separable point by point, we can do pointwise maximization:

$$\arg \max_q \theta q - \frac{q^2}{2} - q \frac{1 - F(\theta)}{f(\theta)}.$$

FOC w.r.t.  $q$ , holding  $\theta$  fixed:

$$\frac{\partial}{\partial q} \left( \theta q - \frac{q^2}{2} - q \frac{1 - F(\theta)}{f(\theta)} \right) = 0.$$

Which gives

$$\theta - q - \frac{1 - F(\theta)}{f(\theta)} = 0.$$

Rearranging for  $q$ :

$$q^{**}(\theta) = \theta - \frac{1 - F(\theta)}{f(\theta)}.$$

Since  $\frac{1 - F(\theta)}{f(\theta)} \geq 0$ ,

$$q^{**}(\theta) \leq \theta = q^*(\theta).$$

There is a downward distortion on quantities for all types below the highest type. For the highest type,  $F(\bar{\theta}) = 1$ , so the distortion term disappears and

$$q^{**}(\bar{\theta}) = \bar{\theta} = q^*(\bar{\theta}).$$

This is the “no distortion at the top” result.

Note: if the hazard rate  $\frac{f(\theta)}{1-F(\theta)}$  is increasing, the monotonicity constraint (i.e.  $q(\theta)$  is increasing) is satisfied.

**Question 6:** Comment. Is allocative efficiency distorted? Are consumers better or worse off compared to the case without private information?

For all types below the top, the quantity is generally distorted downward:

$$q^{**}(\theta) < q^*(\theta)$$

whenever  $\frac{1-F(\theta)}{f(\theta)} > 0$ . The highest type is undistorted. Consumers above the lowest type receive an information rent:

$$U^{**}(\theta) = \int_{\underline{\theta}}^{\theta} q^{**}(x) dx,$$

while the lowest type ( $\underline{\theta}$ ) receives zero. Compared with full information, the monopolist can no longer extract the entire surplus from every type because truthful revelation requires leaving these rents.

**Question 7:** Compare this result with the result of the self-selection exercise with discrete  $\theta$ .

The interpretation is the same. In the discrete problem, the efficient type received an information rent and the inefficient type's allocation was distorted downward: In the continuous problem, the same rent extraction/downward distortion logic applies for every type.

### 3 Optimal Taxation

An economy consists of two types of workers that differ in their productivity:

- $n_1$  workers of productivity  $w_1$ ;
- $n_2$  workers of productivity  $w_2$  with  $w_2 > w_1$ .

Each worker produces the quantity  $Y_i = w_i L_i$ , where  $L_i$  is their labor supply (the number of hours worked). The utility of a worker is increasing in consumption  $C_i$  and decreasing in the number of hours worked  $L_i = Y_i/w_i$ . Utility is given by

$$U_i(C_i, Y_i) = C_i - \left(\frac{Y_i}{w_i}\right)^\alpha,$$

where  $\alpha > 1$  is a fixed parameter for the whole exercise.

**Question 1:** Draw the indifference curves of the workers of type 1 and type 2 in the dimensions  $(Y, C)$ . Verify the single-crossing property.

Fix a utility level  $\bar{U}$ . The indifference curve of type  $i$  is given by

$$U_i(C_i, Y_i) = \bar{U}.$$

Using the utility function,

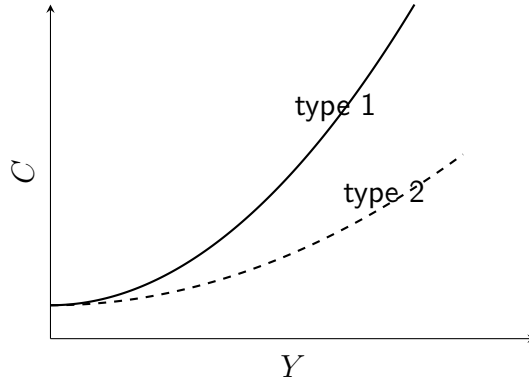
$$C_i - \left(\frac{Y_i}{w_i}\right)^\alpha = \bar{U}.$$

Rearranging,

$$C_i = \bar{U} + \left(\frac{Y_i}{w_i}\right)^\alpha = \bar{U} + \frac{Y_i^\alpha}{w_i^\alpha}.$$

Thus, for a given utility level  $\bar{U}$ , the indifference curves are increasing and convex in the  $(Y, C)$  plane. They all have intercept  $\bar{U}$  when  $Y = 0$ .

For illustration, we can draw the two curves corresponding to the same utility level  $\bar{U}$ , with  $w_2 > w_1$  and  $\alpha > 1$ :



Here, type 1 is the low-productivity type and type 2 the high-productivity type. Since  $w_2 > w_1$ , type 1's indifference curve is steeper.

To get the slope of the indifference curve: Differentiate

$$C_i = \bar{U} + \frac{Y_i^\alpha}{w_i^\alpha}$$

w.r.t  $Y_i$ :

$$\left. \frac{dC_i}{dY_i} \right|_{U_i=\bar{U}} = \frac{\alpha}{w_i^\alpha} Y_i^{\alpha-1}.$$

Notice that this slope does not depend on  $\bar{U}$ . So for a given type, all indifference curves have the same slope at a given  $Y$ .

Now compare the two types. Since  $w_2 > w_1$  and  $\alpha > 1$ ,

$$\frac{1}{w_1^\alpha} > \frac{1}{w_2^\alpha}.$$

Therefore, at any common value of  $Y > 0$ ,

$$\frac{dC_1}{dY} = \frac{\alpha}{w_1^\alpha} Y^{\alpha-1} > \frac{\alpha}{w_2^\alpha} Y^{\alpha-1} = \frac{dC_2}{dY}.$$

So the indifference curve of type 1 is always steeper than that of type 2.

Because the slope of type 1 is always steeper than the slope of type 2 at a given  $Y$ , the two indifference curves can cross at most once. This is exactly the single-crossing property.

Equivalently, if we compare two indifference curves (possibly with different utility levels), their difference is

$$C_1(Y) - C_2(Y) = \bar{U}_1 - \bar{U}_2 + Y^\alpha \left( \frac{1}{w_1^\alpha} - \frac{1}{w_2^\alpha} \right).$$

Since

$$\frac{1}{w_1^\alpha} - \frac{1}{w_2^\alpha} > 0,$$

this difference is strictly increasing in  $Y$ , so it can be equal to zero for at most one value of  $Y$ . Hence the two indifference curves cross at most once.

**Question 2:** Without taxation, each agent consumes all they produce. Determine the labor supply  $L^*(w)$  as a function of productivity. What is the labor supply elasticity? Determine the utility of workers  $U_i^*$  at the optimum.

Without taxes, workers consume all output:

$$C_i = Y_i = w_i L_i.$$

Write the utility after substituting  $C_i = Y_i = w_i L_i$  as  $U_i(L_i) = w_i L_i - L_i^\alpha$ . Worker  $i$  therefore solves

$$\max_{L_i} w_i L_i - L_i^\alpha.$$

The FOC is

$$\frac{dU_i(L_i)}{dL_i} = 0$$

which gives

$$w_i - \alpha L_i^{\alpha-1} = 0,$$

so

$$L_i^* = L^*(w_i) = \left( \frac{w_i}{\alpha} \right)^{\frac{1}{\alpha-1}}.$$

Elasticity of labor supply w.r.t productivity is

$$\varepsilon_{L,w} = \frac{dL^*(w)}{dw} \frac{w}{L^*(w)} = \frac{d \ln L^*(w)}{d \ln w}.$$

We have

$$L^*(w) = \left( \frac{w}{\alpha} \right)^{\frac{1}{\alpha-1}},$$

so

$$\ln L^*(w) = \frac{1}{\alpha-1} (\ln w - \ln \alpha),$$

therefore, differentiating w.r.t  $\ln w$  gives

$$\frac{d \ln L^*}{d \ln w} = \frac{1}{\alpha - 1}.$$

For utility, use the FOC  $w_i = \alpha(L_i^*)^{\alpha-1}$ . Multiplying by  $L_i^*$  gives

$$w_i L_i^* = \alpha(L_i^*)^\alpha.$$

Hence

$$U_i^* = w_i L_i^* - (L_i^*)^\alpha = (\alpha - 1)(L_i^*)^\alpha = (\alpha - 1) \left( \frac{w_i}{\alpha} \right)^{\frac{\alpha}{\alpha-1}}.$$

**Question 3:** The government's objective function is  $n_1 \Psi(U_1) + n_2 \Psi(U_2)$ , where  $\Psi$  is a strictly increasing and strictly concave function that represents the government's preference for redistribution. Suppose that the government observes the type of workers and can implement lump-sum transfers, which is the first-best optimum. Note that transfers:  $T_i > 0$  represent a tax on workers of type  $i$  and  $T_i < 0$  represent a subsidy of agents of type  $i$ . The consumption of agent  $i$  is therefore  $C_i = Y_i - T_i$ . The government's budget constraint is:

$$n_1 T_1 + n_2 T_2 = R.$$

Show that in this case, the government fully redistributes and labor supply is first-best optimal.

A lump-sum tax does not depend on labor. Worker  $i$ 's utility is

$$U_i = w_i L_i - L_i^\alpha - T_i.$$

The FOC remains (since  $T_i$  does not depend on  $L_i$ )

$$w_i - \alpha L_i^{\alpha-1} = 0.$$

Therefore

$$L_i = L^*(w_i).$$

Lump-sum taxation redistributes income without distorting labor supply.

Since labor is first-best, utility can be written

$$U_i = U_i^* - T_i.$$

The government maximizes

$$n_1 \Psi(U_1) + n_2 \Psi(U_2)$$

s.t. the budget constraint. Use the budget constraint to eliminate  $T_1$ :

$$T_1 = \frac{R - n_2 T_2}{n_1}.$$

Then

$$\frac{dU_1}{dT_2} = \frac{n_2}{n_1}, \quad \frac{dU_2}{dT_2} = -1.$$



After substituting the budget constraint, social welfare is a function  $W(T_2)$  of  $T_2$  alone. Differentiating w.r.t  $T_2$ ,

$$\frac{dW(T_2)}{dT_2} = n_2(\Psi'(U_1) - \Psi'(U_2)).$$

At the optimum,

$$\Psi'(U_1) = \Psi'(U_2).$$

Because  $\Psi$  is strictly concave,  $\Psi'$  is strictly decreasing, so equality of marginal social utilities implies

$$U_1 = U_2.$$

**Question 4:** To simplify the analysis, assume now that the government's objective function is  $n_1U_1 + \mu n_2U_2$ , where  $\mu < 1$  reflects the government's preference for low productivity workers. Assume that the government observes neither the type of agents, nor their labor supply (second best optimum). The government only observes the pre-tax revenue  $Y_i = w_iL_i$ . The government will therefore offer agents the choice between two options in a menu  $(T_1, Y_1)$  and  $(T_2, Y_2)$ . In other words, if the government observes the level of production  $Y_i$ , it requires the tax  $T_i$ .

(a) Write the incentive constraints as a function of  $Y_1, Y_2, T_1, T_2, w_1, w_2$ .

(b) Show that the most productive agents produce more than the less productive agents:  $Y_2 \geq Y_1$ .

(In what follows, we assume that the inequality is strict.)

(c) Show that the incentive constraint of the more productive type is binding.

(d) Determine the transfers  $T_1$  and  $T_2$  as a function of  $Y_1$  and  $Y_2$ , as well as the labor supply of type 2 workers. Compare the labor supply of type 1 workers with the first-best optimum (question 3).

Since  $C_i = Y_i - T_i$  and  $L_i = Y_i/w_i$ , the utility of type  $i$  from contract  $(T, Y)$  is

$$Y - T - v(Y/w_i), \quad v(L) = L^\alpha.$$

Therefore type 1 prefers its own contract if

$$Y_1 - T_1 - v\left(\frac{Y_1}{w_1}\right) \geq Y_2 - T_2 - v\left(\frac{Y_2}{w_1}\right) \quad (IC1),$$

and type 2 prefers its own contract if

$$Y_2 - T_2 - v\left(\frac{Y_2}{w_2}\right) \geq Y_1 - T_1 - v\left(\frac{Y_1}{w_2}\right) \quad (IC2).$$

Adding the two incentive constraints (income and tax terms cancel) yields

$$\begin{aligned} & -v\left(\frac{Y_1}{w_1}\right) - v\left(\frac{Y_2}{w_2}\right) \geq -v\left(\frac{Y_2}{w_1}\right) - v\left(\frac{Y_1}{w_2}\right). \\ \Leftrightarrow & v\left(\frac{Y_2}{w_1}\right) + v\left(\frac{Y_1}{w_2}\right) \geq v\left(\frac{Y_1}{w_1}\right) + v\left(\frac{Y_2}{w_2}\right). \end{aligned}$$

Rearranging,

$$v\left(\frac{Y_2}{w_1}\right) - v\left(\frac{Y_1}{w_1}\right) \geq v\left(\frac{Y_2}{w_2}\right) - v\left(\frac{Y_1}{w_2}\right).$$

Since  $v(L) = L^\alpha$ ,

$$Y_2^\alpha \left( \frac{1}{w_1^\alpha} - \frac{1}{w_2^\alpha} \right) \geq Y_1^\alpha \left( \frac{1}{w_1^\alpha} - \frac{1}{w_2^\alpha} \right).$$

Because  $w_2 > w_1$ , the terms in parentheses are positive. Hence

$$Y_2 \geq Y_1.$$

Now suppose  $IC2$  were slack. Keep  $Y_1$  and  $Y_2$  fixed, reduce the low type's tax by  $\eta_1 > 0$  and increase the high type's tax by  $\eta_2 > 0$ :

$$T'_1 = T_1 - \eta_1, \quad T'_2 = T_2 + \eta_2.$$

Choose the changes so that the government budget is unchanged:

$$n_1\eta_1 = n_2\eta_2.$$

This perturbation makes type 1's own contract more attractive and type 2's contract less attractive, so  $IC1$  becomes easier to satisfy. It makes  $IC2$  harder to satisfy, but because  $IC2$  was assumed slack, it remains satisfied for sufficiently small changes. The change in the government's objective is

$$\Delta W = n_1\eta_1 - \mu n_2\eta_2 = (1 - \mu)n_1\eta_1 > 0,$$

because  $\mu < 1$ . Thus a slack  $IC2$  would allow a welfare-improving redistribution from type 2 to type 1, contradicting optimality. Therefore  $IC2$  binds.

Binding  $IC2$  gives

$$Y_2 - T_2 - v\left(\frac{Y_2}{w_2}\right) = Y_1 - T_1 - v\left(\frac{Y_1}{w_2}\right).$$

Rearrange:

$$T_2 - T_1 = Y_2 - Y_1 - \left( v\left(\frac{Y_2}{w_2}\right) - v\left(\frac{Y_1}{w_2}\right) \right).$$

Define

$$D = T_2 - T_1 = Y_2 - Y_1 - \left( v\left(\frac{Y_2}{w_2}\right) - v\left(\frac{Y_1}{w_2}\right) \right).$$

Together with the government budget constraint  $n_1T_1 + n_2T_2 = R$ , we have  $T_2 = T_1 + D$ , so

$$(n_1 + n_2)T_1 + n_2D = R.$$

Therefore

$$T_1 = \frac{R - n_2D}{n_1 + n_2}, \quad T_2 = \frac{R + n_1D}{n_1 + n_2}.$$

Substitute these transfers into the government's objective,

$$W = n_1(Y_1 - v(Y_1/w_1)) + \mu n_2(Y_2 - v(Y_2/w_2)) + \frac{n_1n_2(1 - \mu)}{n_1 + n_2}D.$$

All terms involving  $Y_2$  are a positive coefficient times  $Y_2 - v(\frac{Y_2}{w_2})$ . Hence maximizing w.r.t  $Y_2$  is exactly the first-best problem for type 2. FOC is

$$1 - \frac{\alpha Y_2^{\alpha-1}}{w_2^\alpha} = 0,$$

implying

$$L_2 = L^*(w_2).$$

No distortion at the top.

For type 1, the objective contains the first-best term  $n_1(Y_1 - v(\frac{Y_1}{w_1}))$  plus the screening term

$$\frac{n_1 n_2 (1 - \mu)}{n_1 + n_2} \left( v\left(\frac{Y_1}{w_2}\right) - Y_1 \right).$$

Define  $g(Y_1) = v(\frac{Y_1}{w_2}) - Y_1$ . Then

$$g'(Y_1) = \frac{\alpha Y_1^{\alpha-1}}{w_2^\alpha} - 1.$$

This derivative is zero at the type-2 first-best output  $Y_2^*$  and is negative below  $Y_2^*$ . Since the separating allocation satisfies  $Y_1 < Y_2 = Y_2^*$ , the screening term is decreasing over the relevant range. It therefore pushes the chosen  $Y_1$  below the value that would maximize the first-best term alone. Hence

$$Y_1 < Y_1^*, \quad L_1 < L_1^*.$$

The reason is that the high-productivity worker is tempted to mimic the low-productivity worker. Reducing  $Y_1$  makes the low type bundle less attractive to type 2, which relaxes the screening problem and permits more redistribution. The cost is an efficiency loss for type 1.

To summarize, as in class, the quantity associated with the low type is distorted downward in order to reduce the incentive of the high type to mimic the low type. Specifically, the optimum is implemented by imposing a positive marginal tax rate on workers of the low type, while the marginal tax rate on the high type is zero, so as to guarantee efficiency  $L_2 = L_2$ . This counter-intuitive result comes from a limitation of the model (Stiglitz, 1982), which only considers the intensive margin of labor supply: the disutility of work  $v(L)$  is a continuous function of the number of hours worked. This result has been questioned by recent research, notably by introducing fixed labor force participation costs (costs that are independent of the number of hours worked), meaning these models also study the extensive margin of labor supply.

## 4 Incentives with moral hazard and adverse selection (P 2020-2021)

**Definition: Adverse selection.** The informed party takes advantage of asymmetric information to benefit more from a contract, trade, exchange. One example of that is the health insurance market: the insurer is not informed about the health status of the consumer, and this consumer might be riskier (more costly) than expected.

An investor hires a manager to carry out a project. The project generates a profit of  $R$  in case of success and zero in case of failure. The probability of success is  $e\theta$ , where  $e$  is the effort of the manager and  $\theta$  represents the type of the manager ("talent"). The private cost of effort of the manager is  $\frac{1}{2}ce^2$ . Their outside option is normalized to zero.

If not otherwise mentioned, the investor does not know the type of the manager. The investor only knows that the type can be high ( $\theta = \theta_H$ ) or low ( $\theta = \theta_L$ ), with  $\theta_L < \theta_H$  and that the probability of a manager being of high type is  $\rho \in [0, 1]$ . The investor and the manager are risk neutral. The model parameters  $(c, \theta_L, \theta_H, \rho)$  are known to both actors. A contract specifies a fixed transfer  $t$  paid by the investor to the manager and a transfer contingent on the success of the project,  $r$ . Because the investor is faced with two potential types of managers, they let the manager choose between two types of contracts:  $(t_H, r_H)$  and  $(t_L, r_L)$ . The first option is designed for the high type manager; the second for the low type manager.

**Question 1:** For  $\theta = \theta_L, \theta_H$ , write down the profit of the investor,  $\pi(\theta, e)$ , the utility of the manager,  $U(\theta, e)$ , and the total surplus,  $S(e, \theta)$ , as a function of  $\theta, e, R, r, c$ , and  $t$ . Find the first best level of effort  $e^*(\theta)$  that maximizes the surplus. Under which condition on  $r$  does the manager of type  $\theta$  choose the effort level  $e^*(\theta)$ ?

(Tip: Consider  $\pi$  as the expected profit,  $U$  as the expected utility,  $S$  as the expected surplus.)

If the project succeeds, the investor receives  $R$  and pays the bonus  $r$ , so the investor keeps  $R - r$ . Since success occurs with probability  $\theta e$ , expected profit is

$$\pi(\theta, e) = \theta e(R - r) - t.$$

The manager receives the fixed transfer  $t$ , expected bonus  $\theta e r$ , and pays effort cost  $\frac{1}{2}ce^2$ :

$$U(\theta, e) = t + \theta e r - \frac{1}{2}ce^2.$$

Total surplus is the utility of manager + profit of investor:

$$S(\theta, e) = \pi(\theta, e) + U(\theta, e) = \theta e R - \frac{1}{2}ce^2.$$

Transfers cancel because they only redistribute surplus between investor and manager.

The first-best effort maximizes total surplus:

$$\max_e \theta e R - \frac{1}{2}ce^2.$$

The FOC is

$$\theta R - ce = 0,$$

so

$$e^*(\theta) = \frac{\theta R}{c}.$$

The manager, however, chooses effort to maximize private utility:

$$\max_e t + \theta e r - \frac{1}{2}ce^2.$$

The FOC is

$$\theta r - ce = 0,$$

so

$$e(\theta) = \frac{\theta r}{c}.$$

To implement first-best effort, require  $e(\theta) = e^*(\theta)$ , which gives

$$r = R.$$

When  $r = R$ , the manager internalizes the full return from increasing the success probability.

**Question 2:** Assume in this question, that the investor knows the type of the manager  $\theta$ . Find the optimal contract.

(Tip: Write down the program of the investor and the corresponding constraints before trying to maximize.)

The investor solves

$$\max_{r,t} \theta e(R - r) - t$$

s.t. the manager's effort choice

$$e = \frac{\theta r}{c}$$

and participation constraint

$$t + \theta e r - \frac{1}{2} c e^2 \geq 0.$$

The participation constraint must bind. If the manager received strictly positive utility, the investor could reduce  $t$  slightly, leaving effort unchanged because effort depends on  $r$ , not on  $t$ , and the manager would still participate for a small reduction. Profit would rise one-for-one. Hence

$$t + \theta e r - \frac{1}{2} c e^2 = 0,$$

so

$$t = \frac{1}{2} c e^2 - \theta e r.$$

Substitute this into profit:

$$\pi = \theta e(R - r) - \left( \frac{1}{2} c e^2 - \theta e r \right) = \theta e R - \frac{1}{2} c e^2.$$

Thus, once participation binds, the investor captures total surplus and wants to choose the effort that maximizes it. Using  $e = \theta r / c$ ,

$$\pi(r) = \frac{\theta^2 r R}{c} - \frac{\theta^2 r^2}{2c}.$$

Differentiate:

$$\frac{d\pi}{dr} = \frac{\theta^2}{c} (R - r).$$

Therefore

$$r^* = R.$$

The manager then chooses first-best effort

$$e^*(\theta) = \frac{\theta R}{c}.$$

To find the fixed transfer, use binding participation:

$$t^* = \frac{1}{2}c \left( \frac{\theta R}{c} \right)^2 - \theta \left( \frac{\theta R}{c} \right) R = -\frac{\theta^2 R^2}{2c}.$$

The negative  $t$  means the manager pays the investor upfront. Because  $r = R$ , the manager receives the full project return in the success state and therefore has efficient effort incentives; the negative fixed payment then allows the investor to extract the resulting surplus while leaving the manager exactly at the reservation utility. In the wording of the original correction, the investor effectively “sells” the project to the manager.

Now we assume that the investor does not know  $\theta$ . Let  $U_H$  (respectively  $U_L$ ) be the indirect utility of the manager of type  $H$  (type  $L$ ).

**Question 3:** Show that the incentive constraints can be written as:

$$\frac{r_L^2}{2c} \leq \frac{U_H - U_L}{\theta_H^2 - \theta_L^2} \leq \frac{r_H^2}{2c}.$$

(Tip: Write the incentive constraints by using the effort level chosen by the manager.)

Under a contract  $(t_i, r_i)$ , a type  $\theta$  chooses

$$e(\theta; r_i) = \frac{\theta r_i}{c}.$$

Substituting optimal effort into utility,

$$U(\theta; t_i, r_i) = t_i + \theta \left( \frac{\theta r_i}{c} \right) r_i - \frac{1}{2}c \left( \frac{\theta r_i}{c} \right)^2 = t_i + \frac{\theta^2 r_i^2}{2c}.$$

Therefore truthful utilities are

$$U_H = t_H + \frac{\theta_H^2 r_H^2}{2c}, \quad U_L = t_L + \frac{\theta_L^2 r_L^2}{2c}.$$

The high type must prefer contract  $H$  to contract  $L$ :

$$U_H \geq t_L + \frac{\theta_H^2 r_L^2}{2c}.$$

Subtract  $U_L = t_L + \frac{\theta_L^2 r_L^2}{2c}$ :

$$U_H - U_L \geq \frac{(\theta_H^2 - \theta_L^2) r_L^2}{2c}.$$

Hence

$$\frac{U_H - U_L}{\theta_H^2 - \theta_L^2} \geq \frac{r_L^2}{2c}.$$

Similarly, the low type must prefer contract  $L$  to contract  $H$ :

$$U_L \geq t_H + \frac{\theta_L^2 r_H^2}{2c}.$$

Subtract  $U_H = t_H + \frac{\theta_H^2 r_H^2}{2c}$ :

$$U_L - U_H \geq \frac{(\theta_L^2 - \theta_H^2) r_H^2}{2c}.$$

Multiplying by  $-1$  reverses the inequality:

$$U_H - U_L \leq \frac{(\theta_H^2 - \theta_L^2) r_H^2}{2c}.$$

Therefore

$$\frac{r_L^2}{2c} \leq \frac{U_H - U_L}{\theta_H^2 - \theta_L^2} \leq \frac{r_H^2}{2c}.$$

For nonnegative success payments, this also implies  $r_H \geq r_L$ .

**Question 4:** Express the expected profit of the investor in terms of the total surplus  $S(\theta_L, e_L)$ ,  $S(\theta_H, e_H)$  and the rent left to each type of manager,  $U_H$  and  $U_L$ .

For each type,

$$S = \pi + U,$$

so

$$\pi = S - U.$$

The high type occurs with probability  $\rho$  and the low type with probability  $1 - \rho$ . Therefore

$$\mathbb{E}(\pi) = \rho(S(\theta_H, e_H) - U_H) + (1 - \rho)(S(\theta_L, e_L) - U_L).$$

We can therefore write the investor profit as the surplus minus the rents left to the managers: investor profit equals generated surplus minus the rents left to managers.

**Question 5:** Show that the IRH is automatically satisfied if the other constraints are satisfied. Show that the ICH is binding in the optimum. Write down the rent of the type-H manager  $U_H$  by using  $r_L$ .

The low type's participation constraint is

$$U_L = t_L + \frac{\theta_L^2 r_L^2}{2c} \geq 0.$$

The high type's IC constraint requires

$$U_H \geq t_L + \frac{\theta_H^2 r_L^2}{2c}.$$

Since  $\theta_H > \theta_L$ ,

$$t_L + \frac{\theta_H^2 r_L^2}{2c} \geq t_L + \frac{\theta_L^2 r_L^2}{2c} = U_L.$$

Thus

$$U_H \geq U_L \geq 0,$$

which proves that *IRH* is redundant.

Now suppose *ICH* were slack:

$$U_H > t_L + \frac{\theta_H^2 r_L^2}{2c}.$$

Reduce  $t_H$  by a small  $\varepsilon > 0$ . This lowers  $U_H$  but does not affect the effort choice because effort depends on  $r_H$ , not on  $t_H$ . Since *ICH* was strict, it remains satisfied for small  $\varepsilon$ . *IRL* is unchanged because  $t_L$  is unchanged. The low type's IC constraint becomes easier because the *H*-contract is now less attractive. Finally, the investor pays less whenever type *H* occurs, so expected profit rises. This contradicts optimality. Hence *ICH* binds.

Therefore

$$U_H = t_L + \frac{\theta_H^2 r_L^2}{2c}.$$

Using

$$U_L = t_L + \frac{\theta_L^2 r_L^2}{2c},$$

we obtain

$$U_H = U_L + \frac{(\theta_H^2 - \theta_L^2) r_L^2}{2c}.$$

The high type must receive this extra utility because its greater productivity makes the low type contract more attractive to imitate.

**Question 6:** Show that the ICL can be ignored. Show that the IRL is binding in the optimum. Consider the participation constraints. Find the rent of the type-L manager,  $U_L$ .

From Question 3, incentive compatibility implies  $r_H \geq r_L$ , so

$$r_L^2 - r_H^2 \leq 0.$$

From binding *ICH*,

$$t_H + \frac{\theta_H^2 r_H^2}{2c} = t_L + \frac{\theta_H^2 r_L^2}{2c},$$

which gives

$$t_H - t_L = \frac{\theta_H^2}{2c} (r_L^2 - r_H^2).$$

Because the term in parentheses is nonpositive and  $\theta_H^2 \geq \theta_L^2$ ,

$$\frac{\theta_H^2}{2c} (r_L^2 - r_H^2) \leq \frac{\theta_L^2}{2c} (r_L^2 - r_H^2).$$

Hence

$$t_H - t_L \leq \frac{\theta_L^2}{2c} (r_L^2 - r_H^2).$$



Rearranging,

$$t_L + \frac{\theta_L^2 r_L^2}{2c} \geq t_H + \frac{\theta_L^2 r_H^2}{2c}.$$

This is exactly the low type's incentive constraint. Therefore  $ICL$  is redundant.

Now suppose  $IRL$  were slack, so  $U_L > 0$ . Reduce both fixed transfers by the same small  $\varepsilon > 0$ :

$$t'_L = t_L - \varepsilon, \quad t'_H = t_H - \varepsilon.$$

The difference between the utility from the two contracts remains the same for each type, so both ICs are unchanged. Effort choices are unchanged because  $r_L, r_H$  are unchanged. For sufficiently small  $\varepsilon < U_L$ , the low type still participates; and because  $U_H \geq U_L$ , the high type also participates. The investor pays  $\varepsilon$  less in either state, so expected profit rises. This is not compatible with optimality. Hence

$$U_L = 0.$$

Since

$$U_L = t_L + \frac{\theta_L^2 r_L^2}{2c},$$

we have

$$t_L = -\frac{\theta_L^2 r_L^2}{2c}.$$

Substituting  $U_L = 0$  into the binding  $ICH$  rent formula gives

$$U_H = \frac{(\theta_H^2 - \theta_L^2)r_L^2}{2c} > 0.$$

The low type gets zero rent, while the high type gets a positive information rent.

**Question 7:** Find  $r_H$  in the optimum. Which level of effort does the type-H manager choose?

Using the results above,

$$\mathbb{E}(\pi) = \rho \left( S(\theta_H, e_H) - \frac{r_L^2}{2c}(\theta_H^2 - \theta_L^2) \right) + (1 - \rho)S(\theta_L, e_L).$$

When choosing  $r_H$ , the information-rent term does not depend on  $r_H$ . Thus lowering  $r_H$  would only reduce the high type's effort and reduce surplus, it would not reduce the information rent. The investor therefore chooses  $e_H$  to maximize total surplus:

$$\frac{\partial S(\theta_H, e_H)}{\partial e_H} = \theta_H R - c e_H = 0.$$

Hence

$$e_H^{**} = \frac{\theta_H R}{c} = e_H^*.$$

Since the manager chooses

$$e_H = \frac{\theta_H r_H}{c},$$

implementing this effort requires

$$r_H^{**} = R.$$

Again, there is no distortion at the top.

**Question 8:** Find  $r_L$  in the optimum. Compare the effort chosen by the manager of type-L with the first best effort. To what extent does this exercise illustrate the trade-off between rent extraction and efficiency? Where does the inefficiency come from in this model?

The low type chooses

$$e_L = \frac{\theta_L r_L}{c}.$$

Its total surplus is therefore

$$S(\theta_L, e_L) = \theta_L e_L R - \frac{1}{2} c e_L^2 = \frac{\theta_L^2 r_L R}{c} - \frac{\theta_L^2 r_L^2}{2c}.$$

The high type's information rent is

$$U_H = \frac{r_L^2}{2c} (\theta_H^2 - \theta_L^2).$$

Ignoring terms independent of  $r_L$ , expected profit is

$$(1 - \rho) \left( \frac{\theta_L^2 r_L R}{c} - \frac{\theta_L^2 r_L^2}{2c} \right) - \rho \frac{r_L^2}{2c} (\theta_H^2 - \theta_L^2).$$

Differentiate w.r.t  $r_L$ :

$$(1 - \rho) \left( \frac{\theta_L^2 R}{c} - \frac{\theta_L^2 r_L}{c} \right) - \rho \frac{r_L}{c} (\theta_H^2 - \theta_L^2) = 0.$$

Multiply by  $c$  and factor  $r_L$ :

$$(1 - \rho) \theta_L^2 R = r_L \left( (1 - \rho) \theta_L^2 + \rho (\theta_H^2 - \theta_L^2) \right).$$

Therefore

$$r_L^{**} = \frac{(1 - \rho) \theta_L^2 R}{(1 - \rho) \theta_L^2 + \rho (\theta_H^2 - \theta_L^2)}.$$

Compare this with the first-best level  $r_L^* = R$ :

$$\frac{r_L^{**}}{R} = \frac{(1 - \rho) \theta_L^2}{(1 - \rho) \theta_L^2 + \rho (\theta_H^2 - \theta_L^2)} < 1$$

when  $\rho > 0$  and  $\theta_H > \theta_L$ . Hence

$$r_L^{**} < R = r_L^*.$$

Since effort is increasing in  $r_L$ ,

$$e_L^{**} < e_L^*.$$

The low type manager receives weaker success-contingent incentives and exerts suboptimal effort.

The effort of the type-L manager is suboptimal and the type-L manager receives a lower transfer. Facing uncertainty, the investor incentivizes the type-H manager to reveal themselves by worsening the contract terms for the type-L manager. We find the rent-extraction efficiency trade-off in the in extraction of rent and the provision of incentives: to give the type-H manager optimal (efficient) incentives, and extract a lot of rent from them, the investor needs to give suboptimal (not efficient) incentives to the type-L manager. This is why  $rL < R$ .